

Description and choice data for the domain “Itineraries”

Description of the choice domain 29, Itineraries

The prompt question and the universe of five response options in the choice domain **Itineraries** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

This domain illustrates three-way tradeoffs. The points form a constellation in the simplex that resembles the pattern of points on the “five” side of a die. The name for this domain used in data files and documents, “Itineraries” is an abbreviation of the title seen by participants, “Two-flight itineraries”.

Which one of the following flight itineraries would you choose? All involve two flights and have a total duration of six hours.

1st flight	Layover	2nd flight
1:30	1:15	3:15
3:15	1:15	1:30
2:15	1:30	2:15
1:30	1:45	2:45
2:45	1:45	1:30

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

Two-Flight Itineraries

Which one of the following flight itineraries would you choose? All involve two flights and have a total duration of six hours.

	1st flight	Layover	2nd flight
☐	2:15	1:30	2:15
☐	1:30	1:15	3:15
☐	1:30	1:45	2:45
☐	2:45	1:45	1:30

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Figure 1: Screenshot for domain Itineraries

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	18	22	-	-	-	0.450	0.550	-	-	-
$\{a, c\}$	13	-	27	-	-	0.325	-	0.675	-	-
$\{b, c\}$	-	19	21	-	-	-	0.475	0.525	-	-
$\{a, b, c\}$	8	16	16	-	-	0.200	0.400	0.400	-	-
$\{a, d\}$	18	-	-	23	-	0.439	-	-	0.561	-
$\{b, d\}$	-	19	-	21	-	-	0.475	-	0.525	-
$\{a, b, d\}$	10	14	-	16	-	0.250	0.350	-	0.400	-
$\{c, d\}$	-	-	21	19	-	-	-	0.525	0.475	-
$\{a, c, d\}$	12	-	19	9	-	0.300	-	0.475	0.225	-
$\{b, c, d\}$	-	15	10	15	-	-	0.375	0.250	0.375	-
$\{a, b, c, d\}$	8	11	16	5	-	0.200	0.275	0.400	0.125	-
$\{a, e\}$	21	-	-	-	19	0.525	-	-	-	0.475
$\{b, e\}$	-	16	-	-	24	-	0.400	-	-	0.600
$\{a, b, e\}$	6	18	-	-	16	0.150	0.450	-	-	0.400
$\{c, e\}$	-	-	17	-	23	-	-	0.425	-	0.575
$\{a, c, e\}$	15	-	12	-	14	0.366	-	0.293	-	0.341
$\{b, c, e\}$	-	20	11	-	9	-	0.500	0.275	-	0.225
$\{a, b, c, e\}$	8	10	14	-	8	0.200	0.250	0.350	-	0.200
$\{d, e\}$	-	-	-	14	26	-	-	-	0.350	0.650
$\{a, d, e\}$	12	-	-	10	18	0.300	-	-	0.250	0.450
$\{b, d, e\}$	-	20	-	6	14	-	0.500	-	0.150	0.350
$\{a, b, d, e\}$	8	17	-	7	8	0.200	0.425	-	0.175	0.200
$\{c, d, e\}$	-	-	16	7	17	-	-	0.400	0.175	0.425
$\{a, c, d, e\}$	10	-	12	0	18	0.250	-	0.300	0.000	0.450
$\{b, c, d, e\}$	-	15	15	4	6	-	0.375	0.375	0.100	0.150
$\{a, b, c, d, e\}$	8	8	10	5	9	0.200	0.200	0.250	0.125	0.225

Table 1: Observed choice counts and proportions.

Choice data for domain 29, Itineraries

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

